已镜像和未镜像聚合

ONTAP 9

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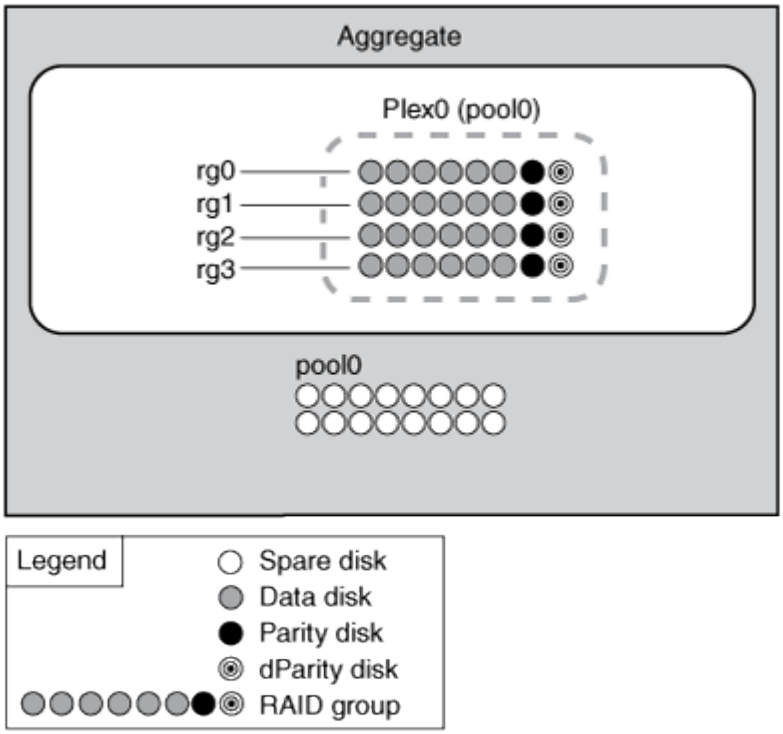
已镜像和未镜像聚合

您可以使用名为 *raid* 的可选功能同步镜像存储在不同 SyncMirror 组中的副本或 *plexes* 中的聚合数据。丛可确保在出现故障的磁盘数量超过 RAID 类型所能保护的磁盘数量或与 RAID 组磁盘的连接断开时防止数据丢失。

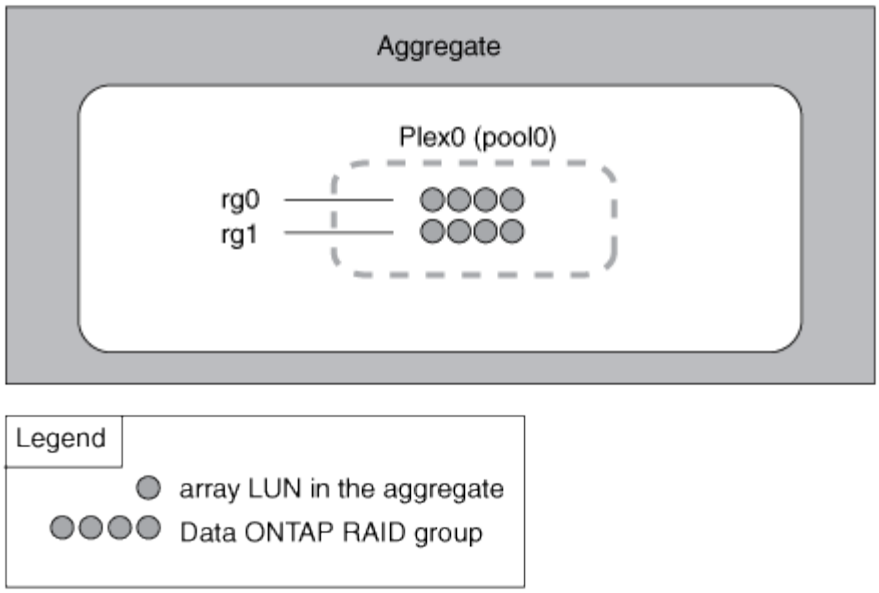
未镜像聚合的工作原理

除非您使用的是 SyncMirror ，否则所有聚合都会取消镜像。未镜像聚合只有一个 *plex* （其数据的副本），其中包含属于该聚合的所有 RAID 组。

下图显示了一个由磁盘组成的未镜像聚合及其一个丛。聚合包含四个 RAID 组： rg0 ， rg1 ， rg2 和 rg3 。每个 RAID 组都有 6 个数据磁盘，一个奇偶校验磁盘和一个双奇偶校验磁盘。聚合使用的所有磁盘都来自同一个池 pool0 。



下图显示了一个包含阵列 LUN 的未镜像聚合及其一个丛。它具有两个 RAID 组： rg0 和 rg1 。聚合使用的所有阵列 LUN 都来自同一个池 pool0 。



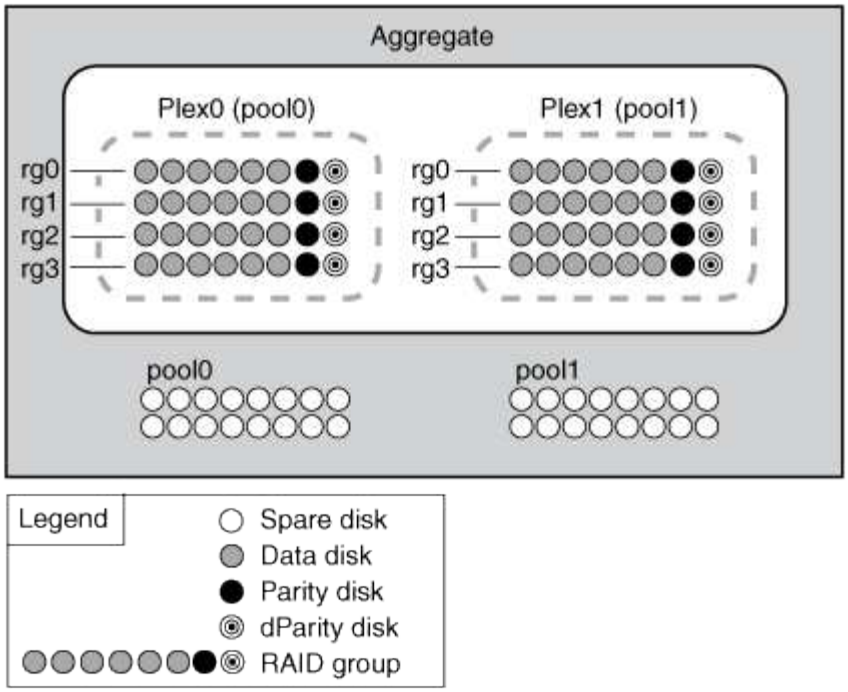
镜像聚合的工作原理

镜像聚合具有两个 *plexes* （其数据的副本），这两个聚合使用 SyncMirror 功能复制数据以提供冗余。

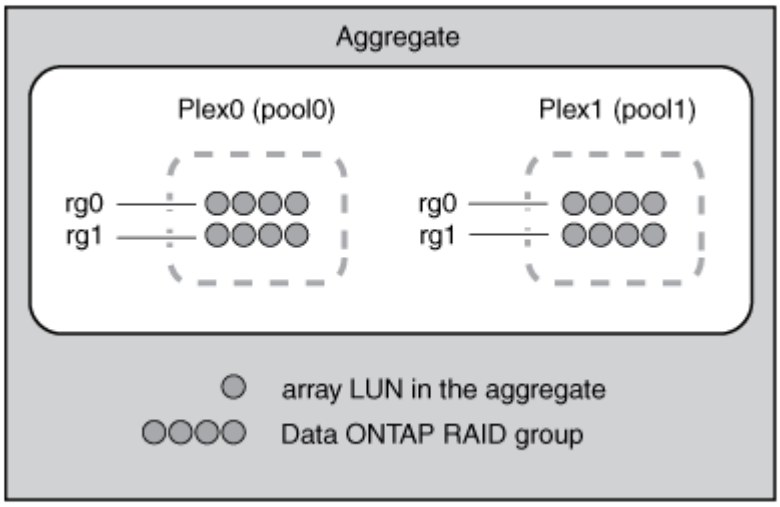
创建镜像聚合（或将第二个丛添加到现有未镜像聚合）时， ONTAP 会将原始丛（ plex0 ）中的数据复制到新丛（ plex1 ）中。丛在物理上是分开的（每个丛都有自己的 RAID 组和池），并且丛会同时更新。这样，如果出现故障的磁盘数超过聚合所保护的 RAID 级别或连接断开，则可以提供额外的保护，防止数据丢失，因为在修复故障发生原因时，未受影响的丛会继续提供数据。修复出现问题的丛后，两个丛将重新同步并重新建立镜像关系。

系统上的磁盘和阵列 LUN 分为两个池： pool0 和 pool1 。Plex0 从 pool0 获取其存储，而 Plex1 从 pool1 获取其存储。

下图显示了由启用和实施了 SyncMirror 功能的磁盘组成的聚合。已为聚合创建第二个丛，即 plex1 。plex1 中的数据是 plex0 中的数据副本， RAID 组也是相同的。32 个备用磁盘分配给 pool0 或 pool1 ，每个池有 16 个磁盘。



下图显示了由启用和实施了 SyncMirror 功能的阵列 LUN 组成的聚合。已为聚合创建第二个丛，即 plex1 。Plex1 是 plex0 的副本， RAID 组也相同。



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